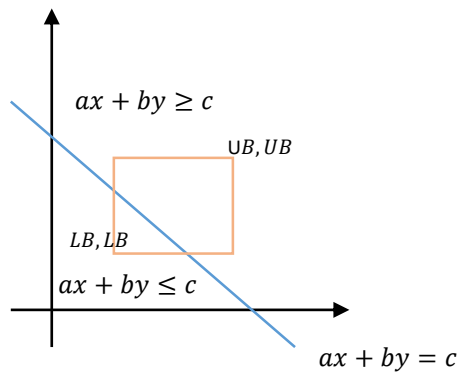


In high school, we have learned a lot about two dimensional plane with x axis and y axis. In 2D plane, a line can be represented by three parameters a , b , and c , forming the equation $ax + by = c$. The area below the line can be expressed as $ax + by \leq c$, while the area above the line is $ax + by \geq c$. As the ranges of x axis and y axis are from negative infinity to positive infinity, the size of these two areas are also infinity. However, if we bound x axis and y axis to a range from LB to UB , we can calculate the square area below or above the line as the figure shown below.



To approximate the area, a simple way is to count the number of discrete points which are beneath or beyond the line. This can be controlled by a directional coefficient $d \in \{-1,1\}$ as a multiplier of a , b , and c . Write a program to help the calculation of such area.

Input

The input contains several cases and ends with all zeros. Each case contains six integer values, which in turn represent LB , UB , a , b , c and d .

Output

For each case, output the number of points satisfied $LB \leq x \leq UB$, $LB \leq y \leq UB$, and $adx + bdy \leq cd$ for all $x, y \in \mathbb{Z}$. The output format is given in sample output. Note that the directional coefficient should only affect the direction of the relation ($\geq$ or $\leq$).

Sample Input

```
0 10 2 3 4 1
0 10 2 3 4 -1
0 10 -2 -3 -4 1
0 10 -2 -3 -4 -1
0 0 0 0 0
```

Sample Output

The square area within 0 to 10 satisfying $2x+3y\leq 4$ is 4.
The square area within 0 to 10 satisfying $2x+3y\geq 4$ is 118.
The square area within 0 to 10 satisfying $-2x-3y\leq -4$ is 118.
The square area within 0 to 10 satisfying $-2x-3y\geq -4$ is 4.